SLA OR NAY?

CAUSAL IMPACTS OF BRISTOL DLAS

Levi John Wolf
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WHAT & WHY ARE DLA?

IDENTIFICATION & ESTIMATION

PRELIMINARY RESULTS

SLA OR NAY?

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PRELIMINARY RESULTS

SLA OR NAY?

DISCRETIONARY LICENSING AUTHORITY

requirements for privately rented properties falling within a council's jurisdiction that extend national requirements for licensing

DISCRETIONARY
LICENSING
AUTHORITY

ADDITIONAL
LICENSING

extends national HMO requirements to "smaller" households of 3 or more people w/ shared facilities

DISCRETIONARY LICENSING AUTHORITY

ADDITIONAL LICENSING

SELECTIVE LICENSING

extends national HMO requirements to all properties for 5 years

DISCRETIONARY LICENSING

DISCRETIONARY LICENSING AUTHORITY

applies

ADDITIONAL LICENSING

to certain kinds of properties

SELECTIVE LICENSING

in specific areas & times

EASTERN SLA (1)

Addressing severe quality issues
2016-2021 in St. George West & Eastville
All privately-let properties
EPC, Electric, Gas safety inspections
£300 to £750 fee

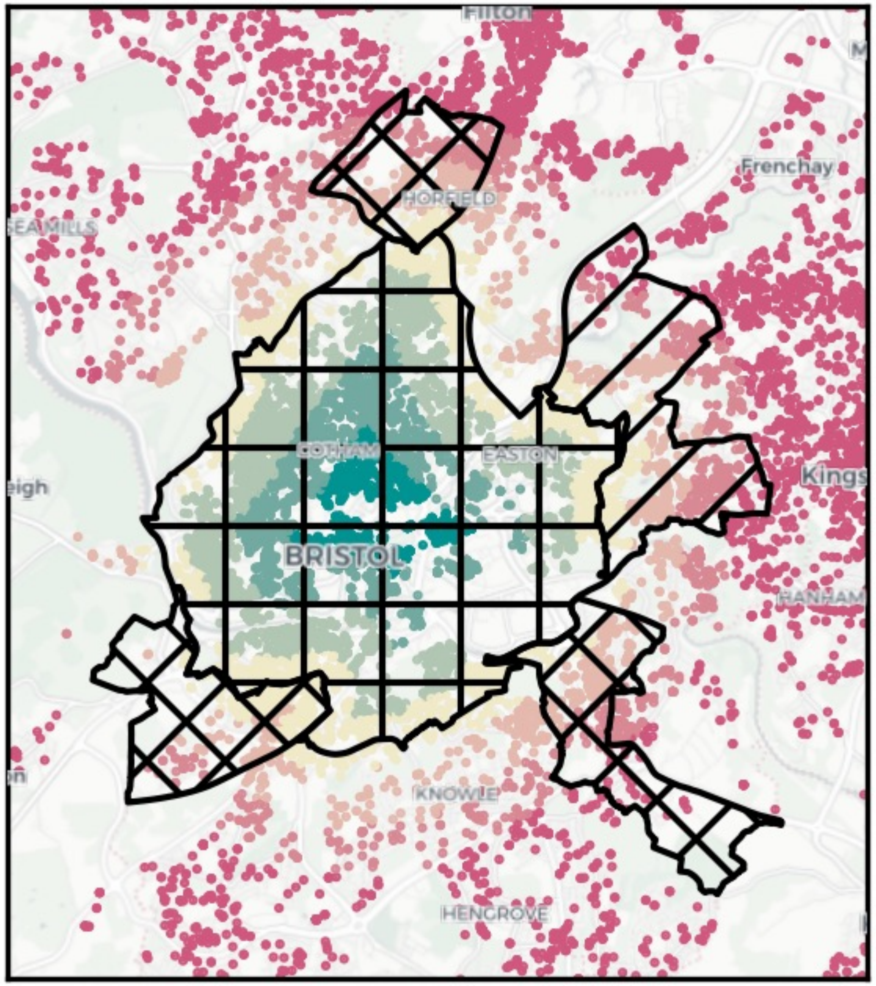
CENTRAL ALA (2)

2019-2024 in all central wards
3+ People w/ Shared Facilities
EPC, Electric, Gas safety inspections
£900 to £1450 fee

EXTENSION (3)

2024-2029 for different targets
EPC, Electric, Gas safety inspections
£900 to £2000 fee

Listings near DLAs



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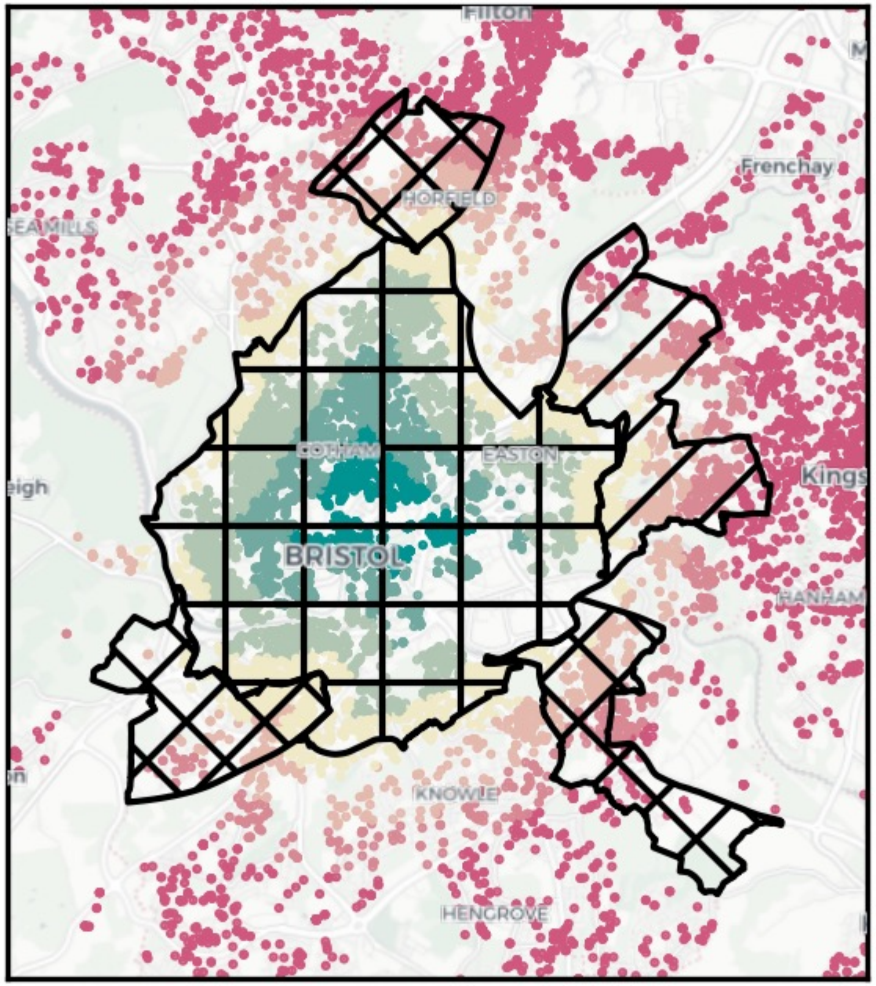

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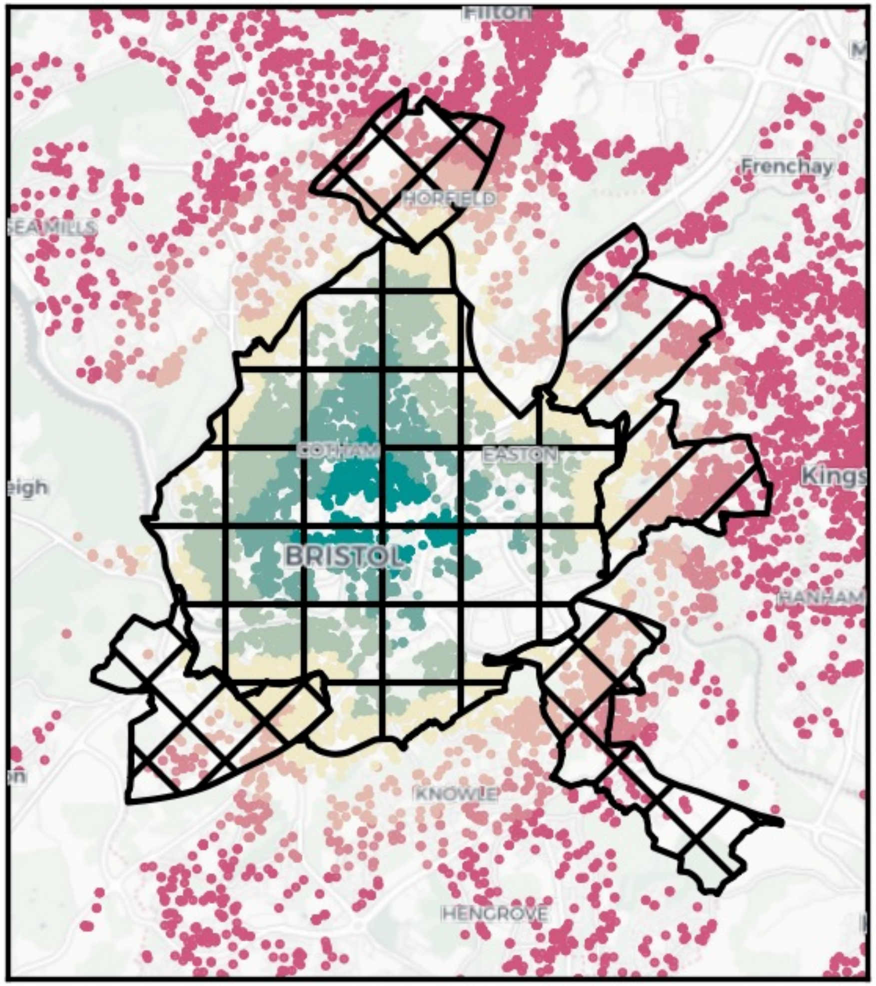

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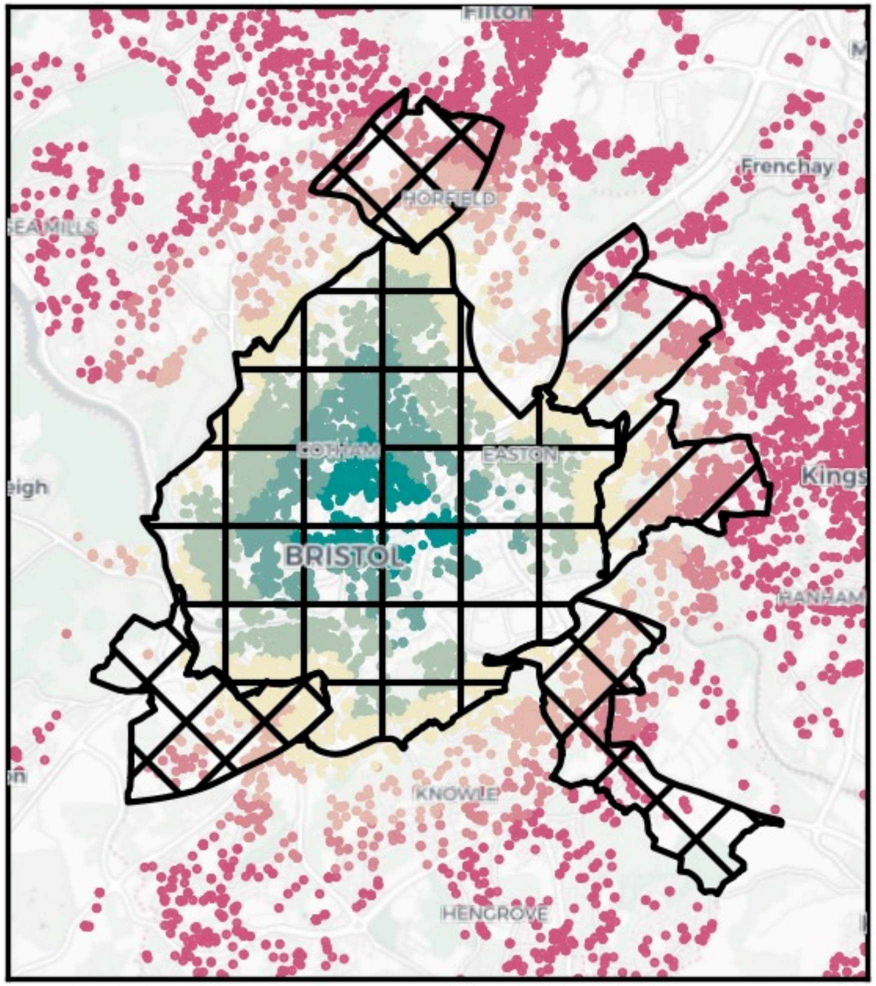

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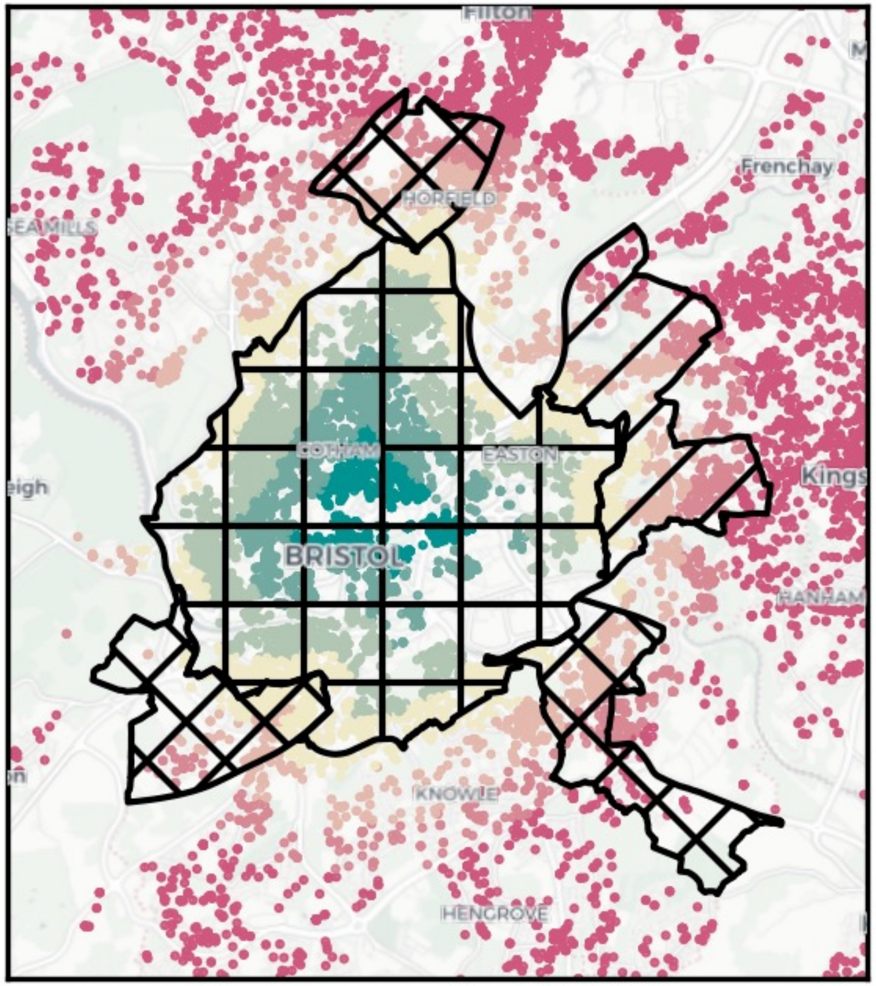

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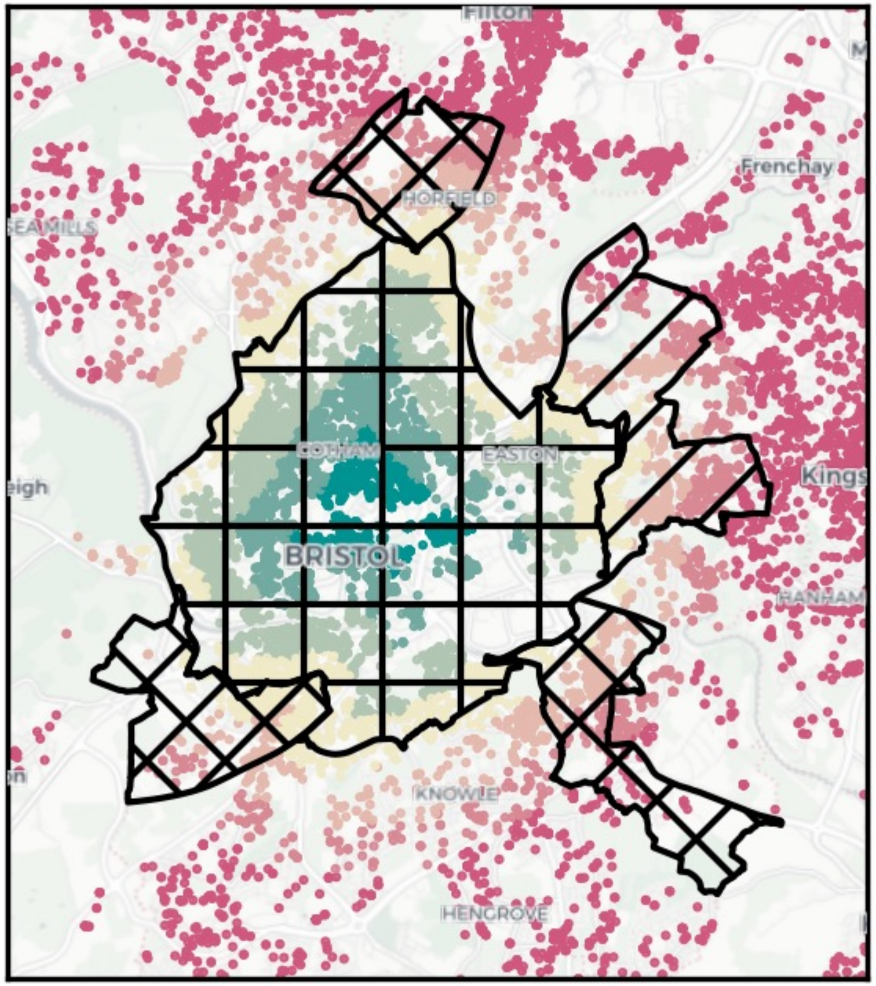

City Council of Bristol (The council)
Housing Act 2004 (“The Act”): Parts 2 and 3
NOTICE OF DESIGNATION OF A CITYWIDE ADDITIONAL LICENSING
NOTICE OF DESIGNATION OF BISHOPSTON AND ASHLEY DOWN, COTHAM AND EASTON WARDS FOR SELECTIVE LICENSING

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BBC

Bristol landlords oppose new licensing fee scheme

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BBC

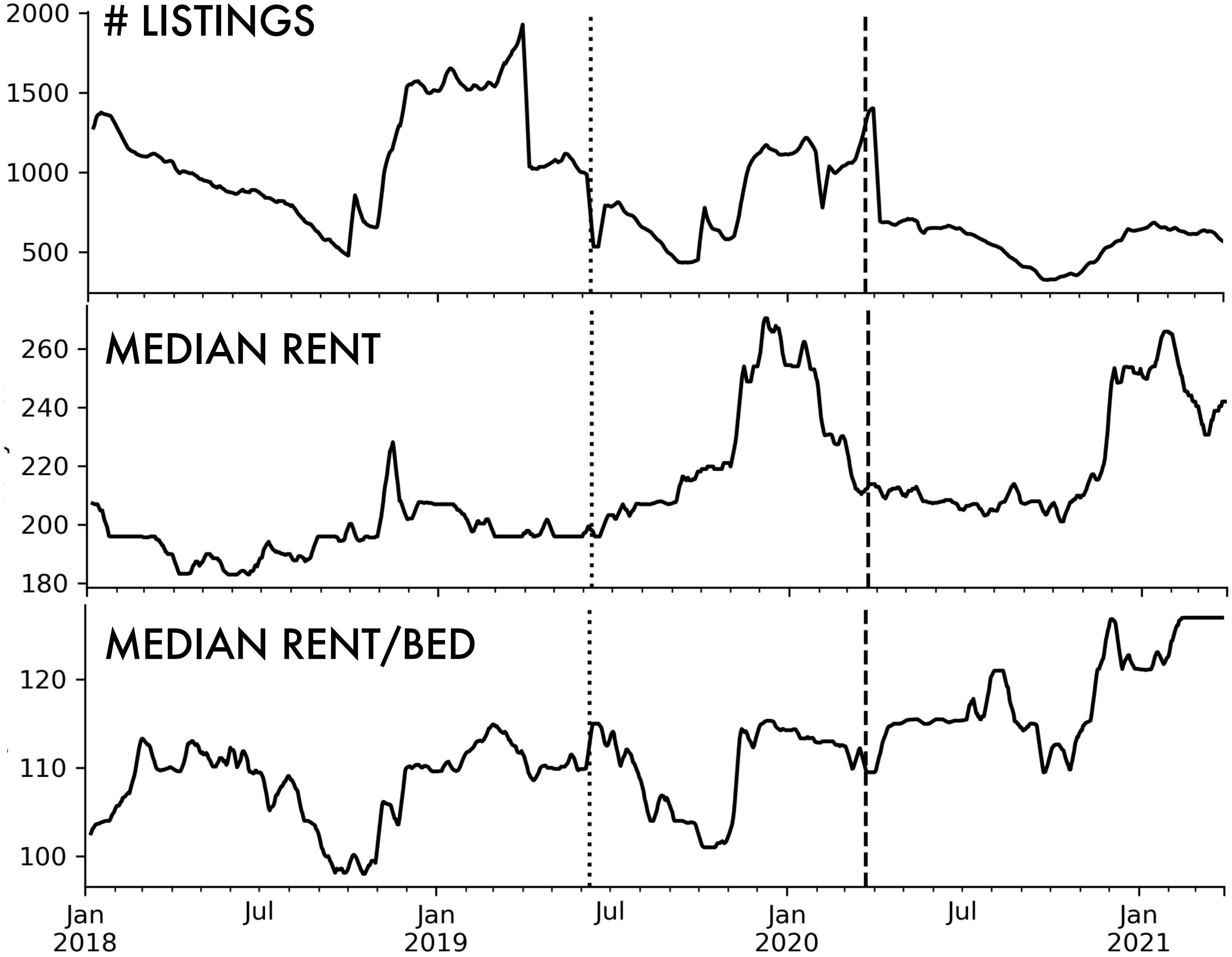
Bristol landlords oppose new licensing fee scheme

Letting agents and landlords criticised the cost of the licences, and also warned the extra fees could lead to higher rents for tenants, less affordable housing getting built, and some landlords leaving the market, meaning fewer homes available to rent.

WHAT & WHY ARE DLA?

IDENTIFICATION & ESTIMATION

PRELIMINARY RESULTS

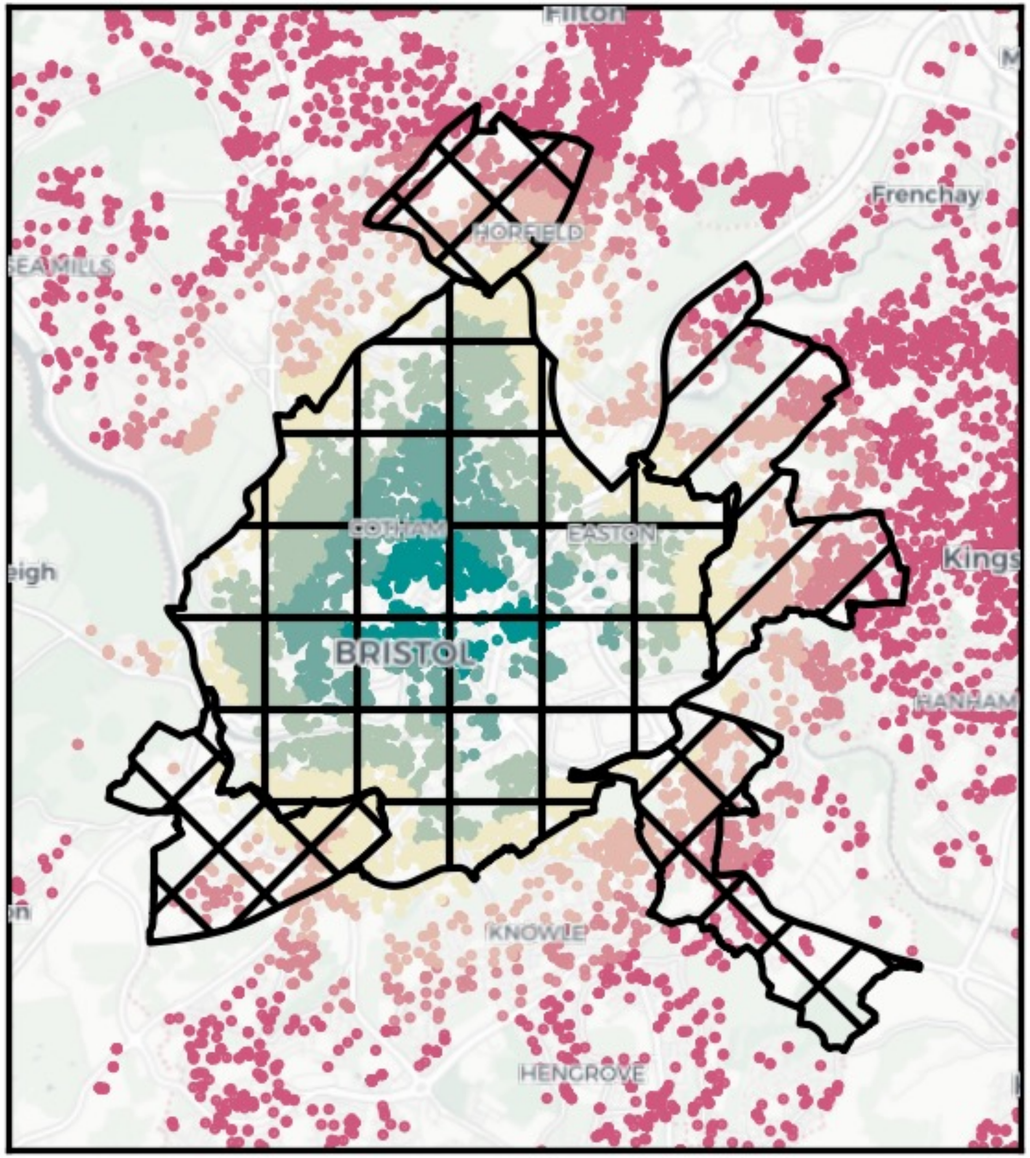


Fewer rentals?

Higher rent?

Less affordable?

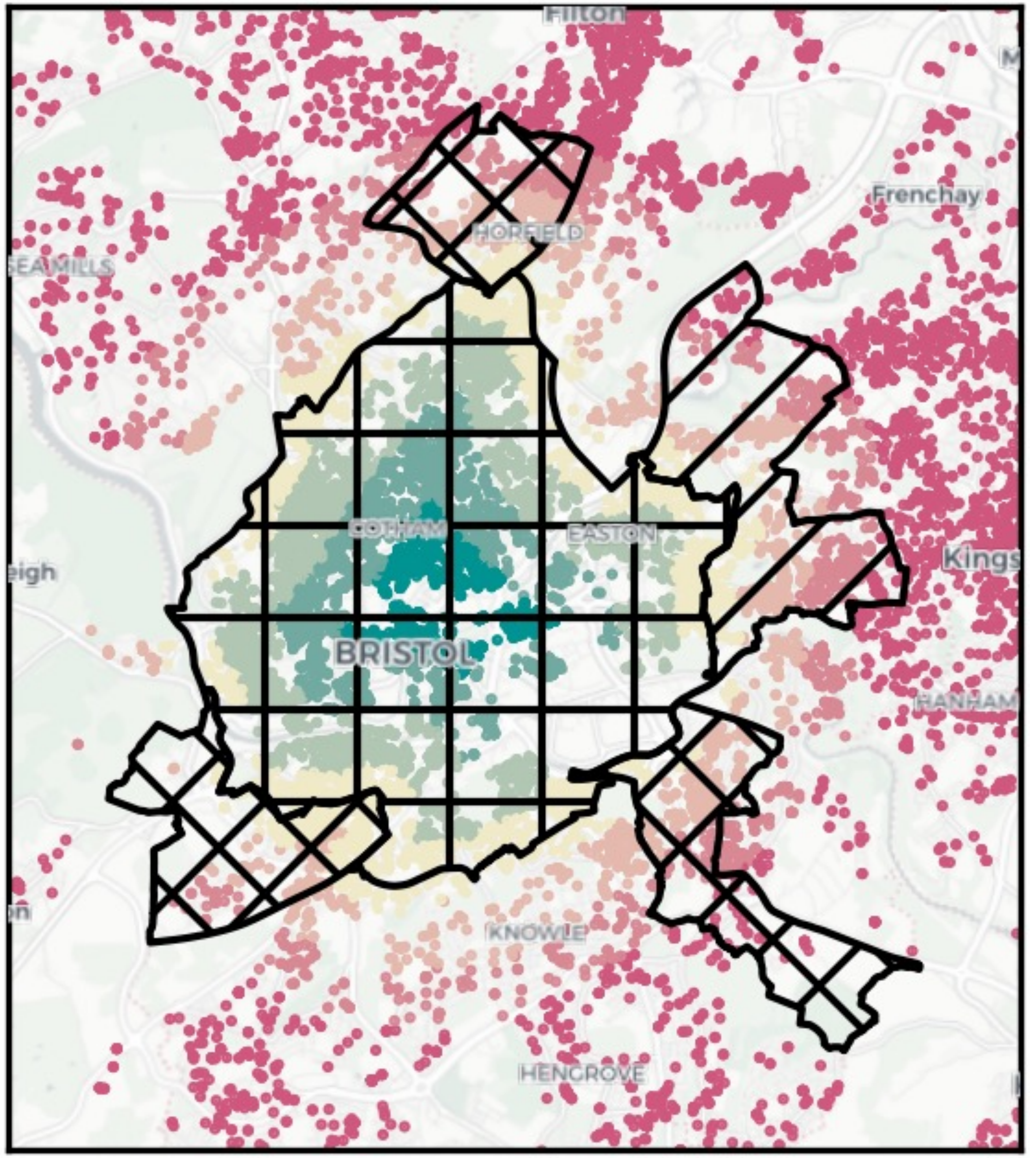
Listings near DLAs



Given the geography of these regulation areas, how can we be sure that their imposition is reflected in the "total" city rates?

Displacements **within** cities matter!

Listings near DLAs



Given the geography of these regulation areas, how can we be sure that their imposition is reflected in the "total" city rates?

Displacements **within** cities matter!

Neighborhood-level rent regulation distorts *local*, not city-wide, incentives.

We want to know: for similar properties, did licensing make a difference *on the margin?*

That is, **at the boundary of the DLA,** can we detect the difference the DLA makes?

Treated listings inside the DLA after treatment.
Pre-treated listings inside the DLA before treatment.
Substitute listings outside the DLA but nearby.
Pre-Substitute listings outside of the DLA before treatment.

IDENTIFICATION STRATEGY 1: SDIDc

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At the DLA boundary, estimate discontinuity
Pre-treated and **Pre-substitute**, then **Treated** and **Substitute**

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If the estimated discontinuity at the DLA boundary *changes*,
we have **a spatial difference in discontinuities.**

ATE: difference in estimated boundary discontinuities

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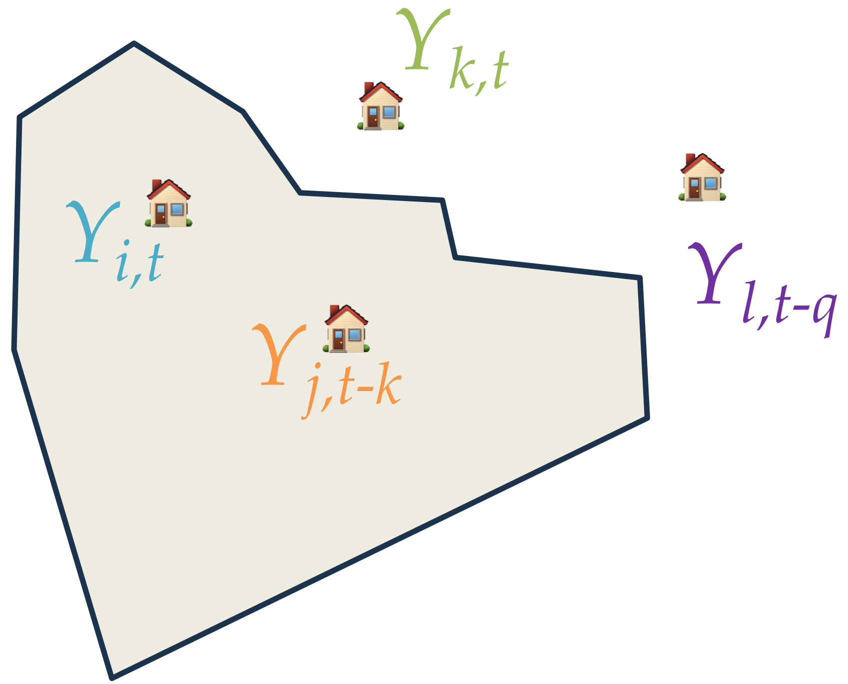

Borrow counterfactual from *just over* the boundary

Pretreatment model:
fit on Y_j POs from Y_l

Treatment model:
fit on Y_i POs from Y_k

Substitute model:
fit on Y_k POs from Y_i

Pre-substitute model:
fit on Y_l POs from Y_j

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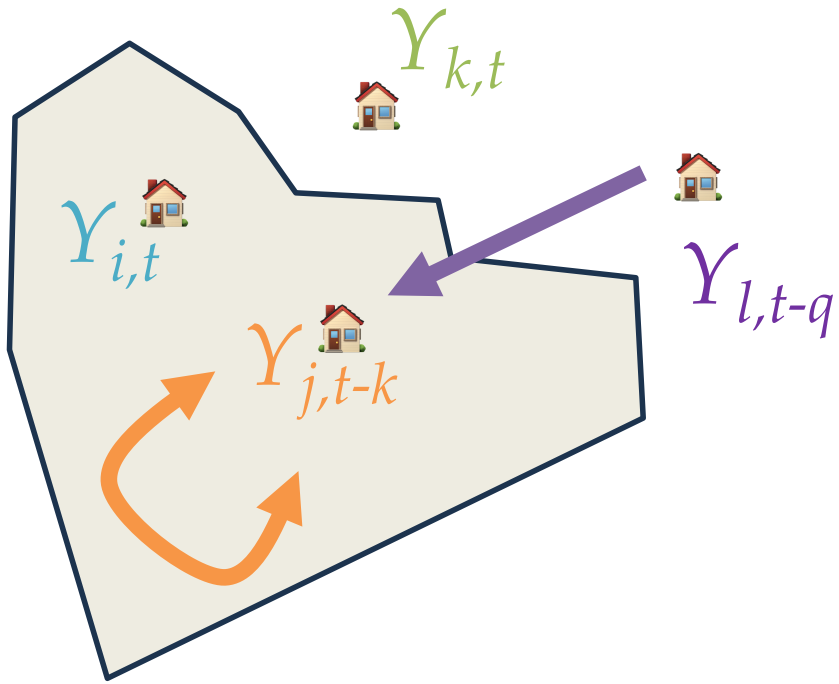

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Discontinuity: predicted outcome under “pre-treated” inside model vs. “never-treated” outside model

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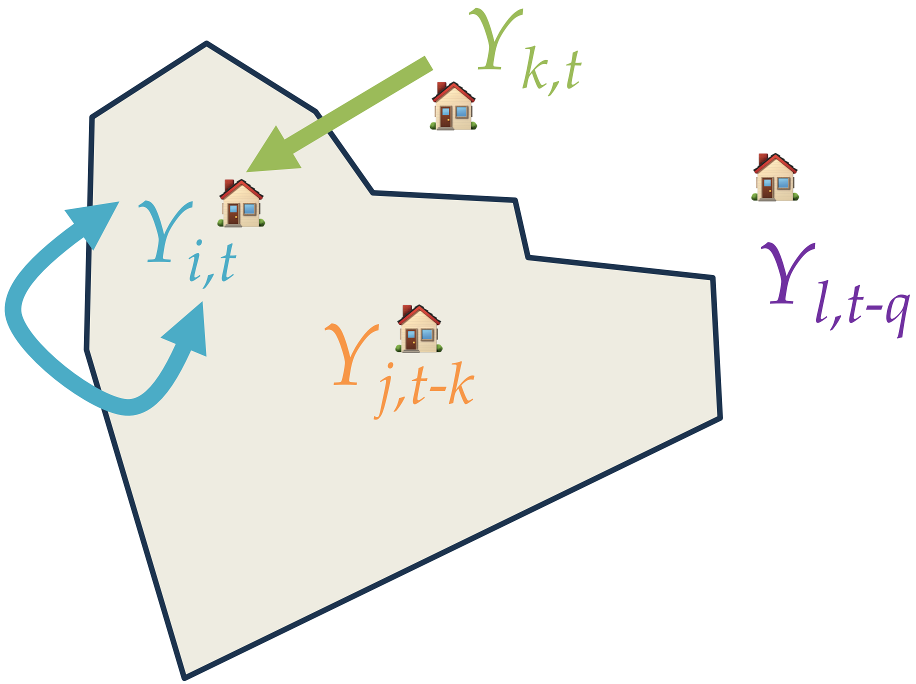

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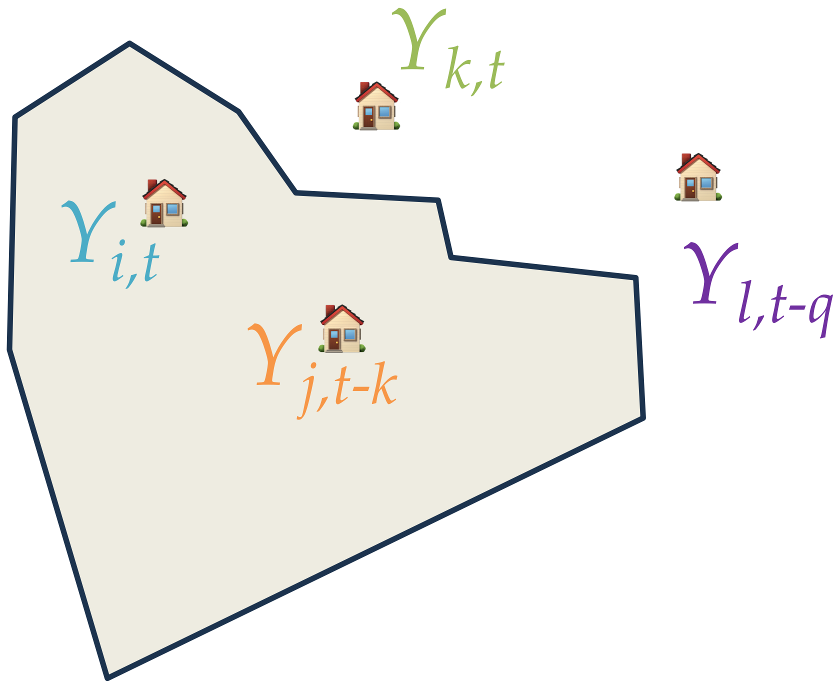

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ATT: average discont. for top 2
ATE: average discont. overall

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IDENTIFICATION STRATEGY 2: QUARTET.+ X

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Match:

1. treated to predecessor—match back in time w/in DLA
2. treated to substitute—match contemporary to out of DLA
3. substitute to predecessor—match back in time out of DLA

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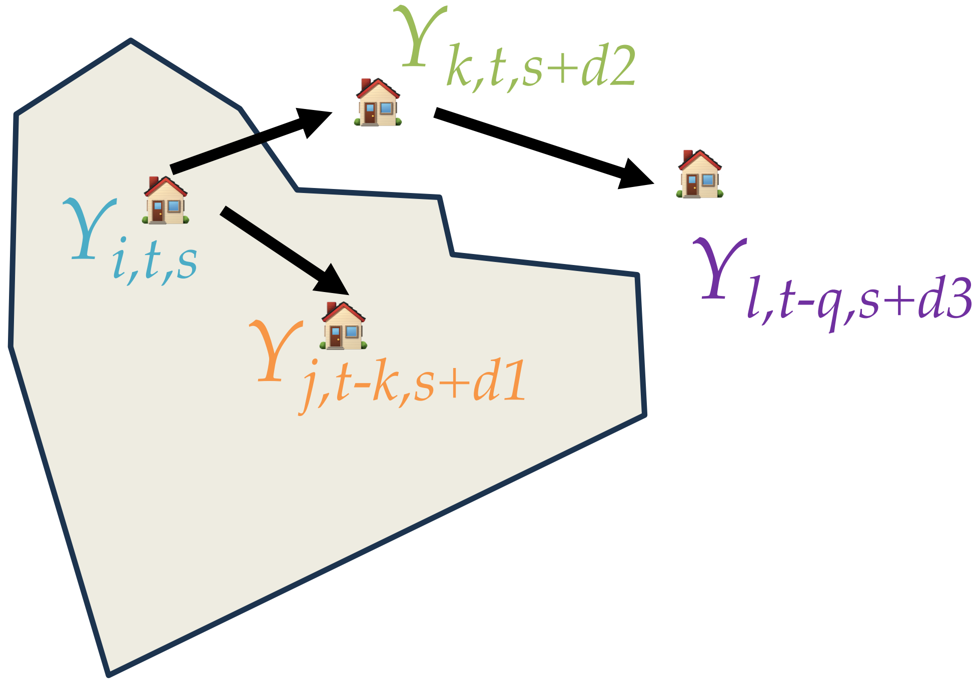

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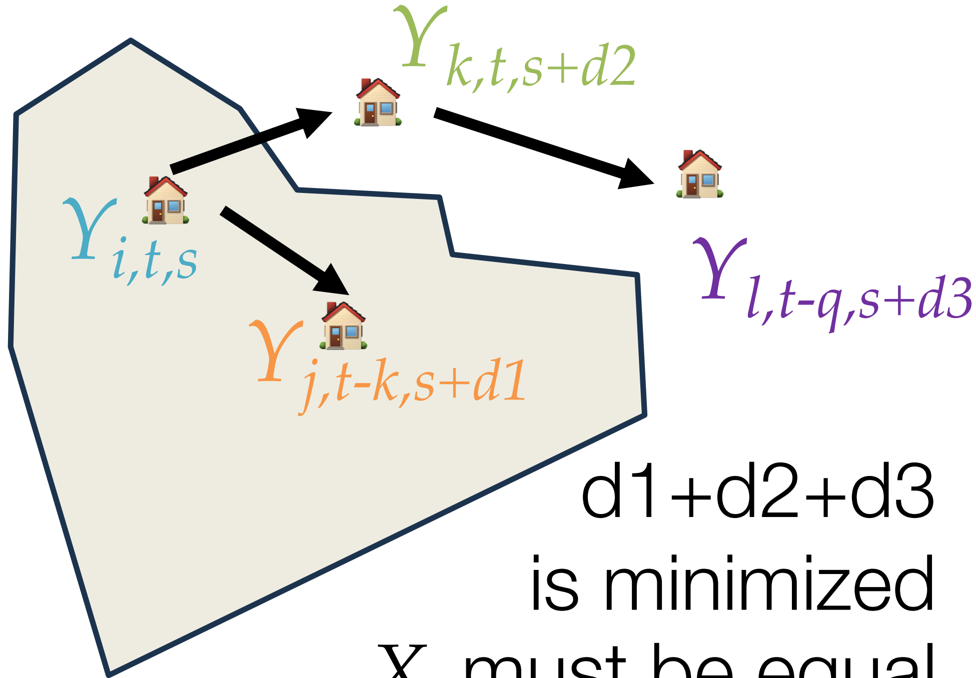

d1+d2+d3
is minimized
$X_{.}$ must be equal

IDENTIFICATION STRATEGY 2: QUARTET + X

$Y_i = Y_h + X_i + e_i$ TREATED AREA MODEL

$Y_k = Y_l + X_k + e_k$ UNTREATED AREA MODEL

$\tau_1 = \hat{Y}_0 - Y_1 = Y_h + X + u$ TREATED PO MODEL

$\tau_0 = Y_0 - \hat{Y}_1 = Y_l + X + u$ UNTREATED PO MODEL

$t \sim Y + X$ PROPENSITY MODEL

ATE: $n^{-1} \Sigma\ \tau_1 p + \tau_0 (1-p)$

IDENTIFICATION STRATEGY 2: QUARTET + X

All models are GWRFs using rental characteristics (about 125 nearest neighbors; X-learner propensity model uses 1000-meter fixed bandwidth)

All inference uses residual bootstrap

Additionally, for X-learner we also shuffle the propensity score separately

ESTIMATION & INFERENCE

WHAT & WHY ARE DLA?

IDENTIFICATION & ESTIMATION

PRELIMINARY RESULTS

SLA OR NAY?

kind	uid	Treated	In DLA?	Date	Postcode	Beds	Baths	Rent
treated	9cae7e1edb	TRUE	TRUE	2019-08-14	BS34AP	2	0	242
pretreat	6f91be8c67	FALSE	TRUE	2018-08-16	BS34AP	2	0	230
substitute	573f5484f2	FALSE	FALSE	2019-06-03	BS44JD	2	0	207
presubstitute	70cfae8c52	FALSE	FALSE	2018-01-07	BS44AU	2	0	167

MATCH EXAMPLE

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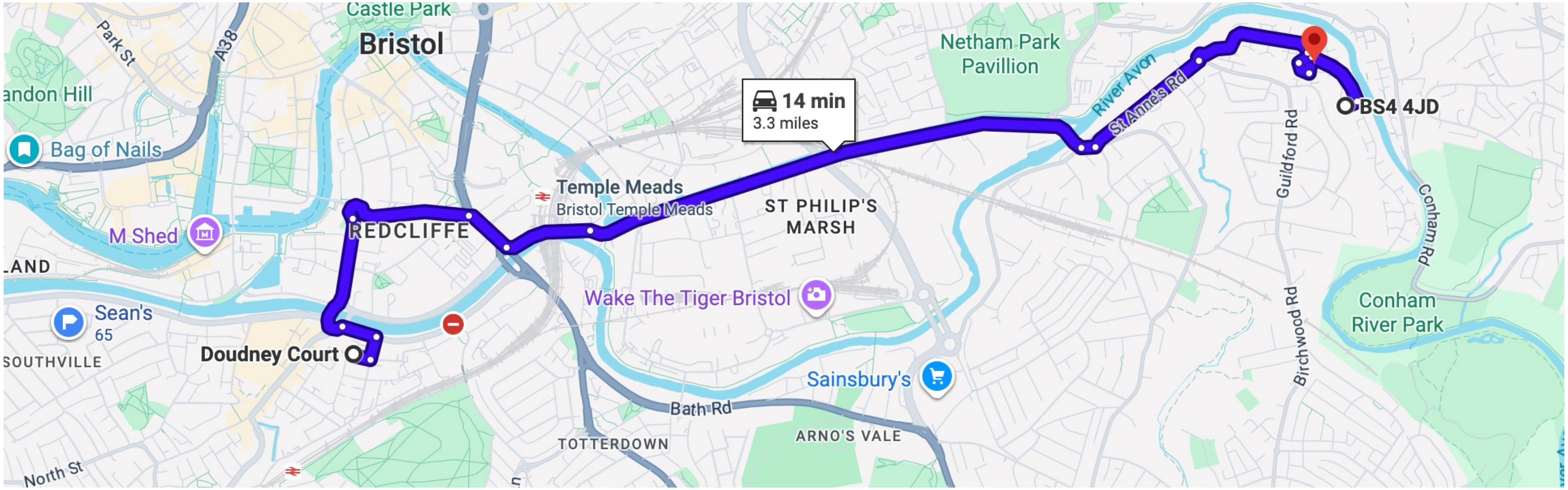

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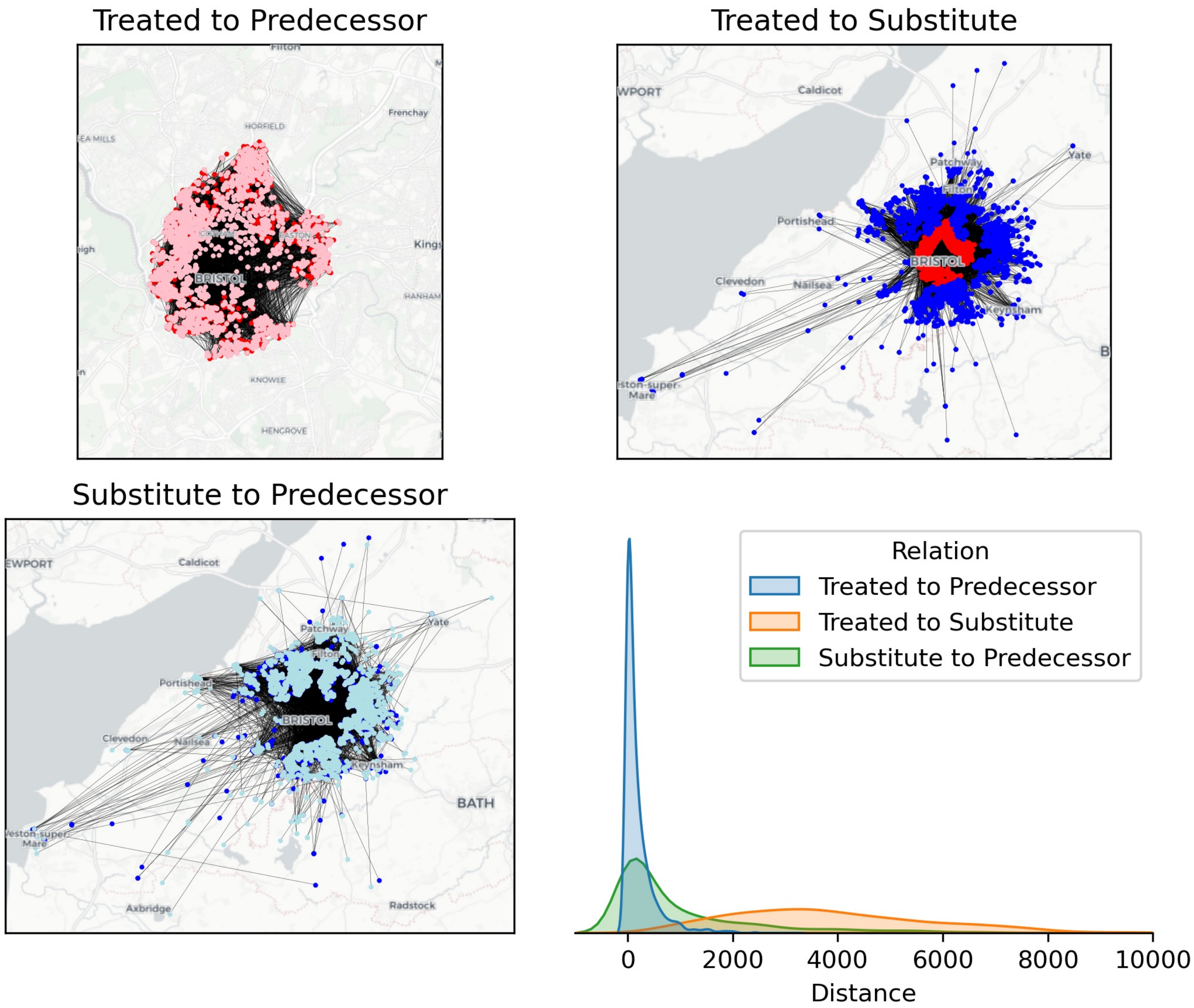


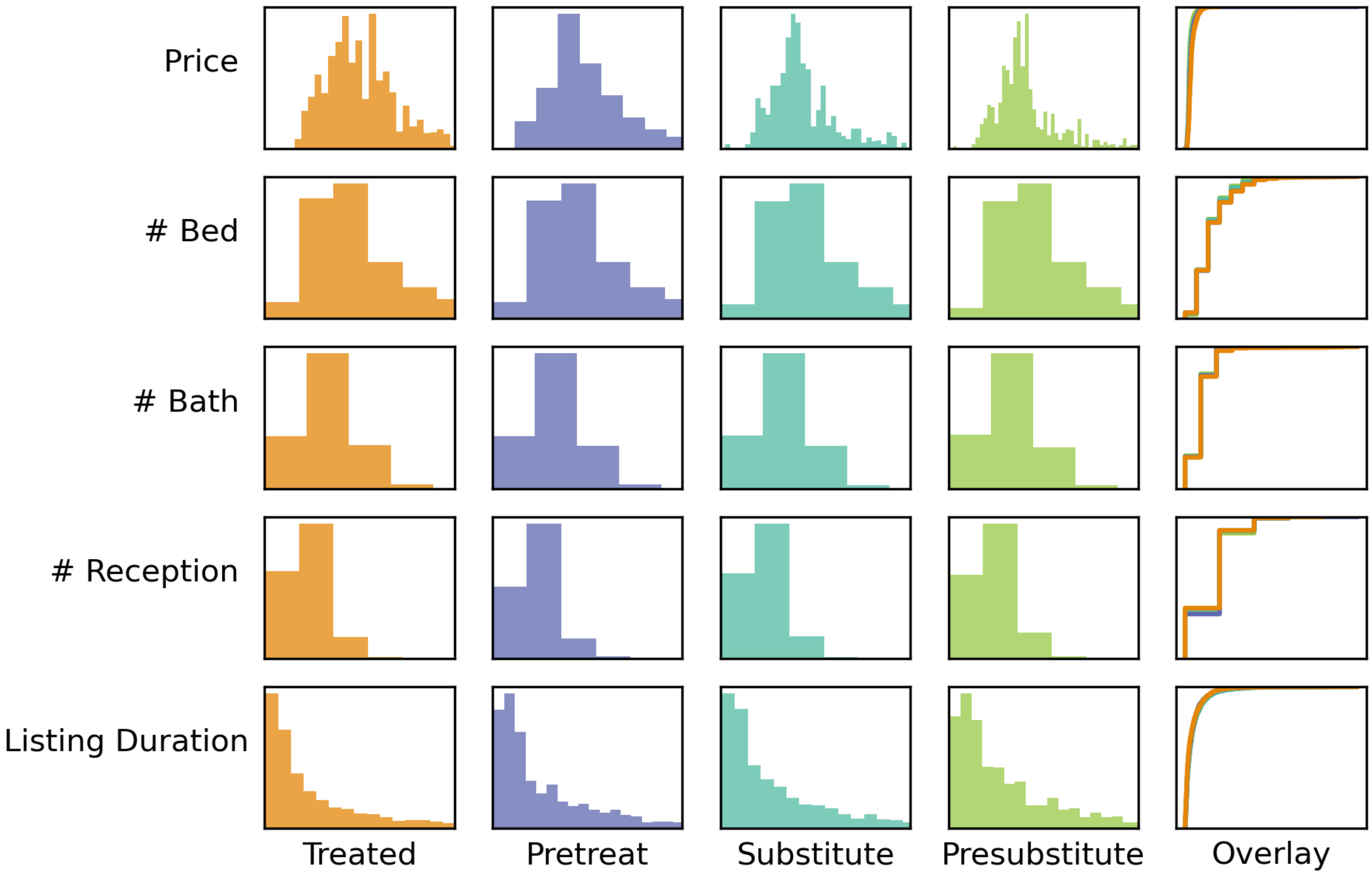
Proximity of the step-wise matching leaves something to be desired… but!

listings are 2018-2022
policy is 2019-2024

2 more years of data for substitutes won't necessarily improve this a large amount.

Computation is still running for the joint match.

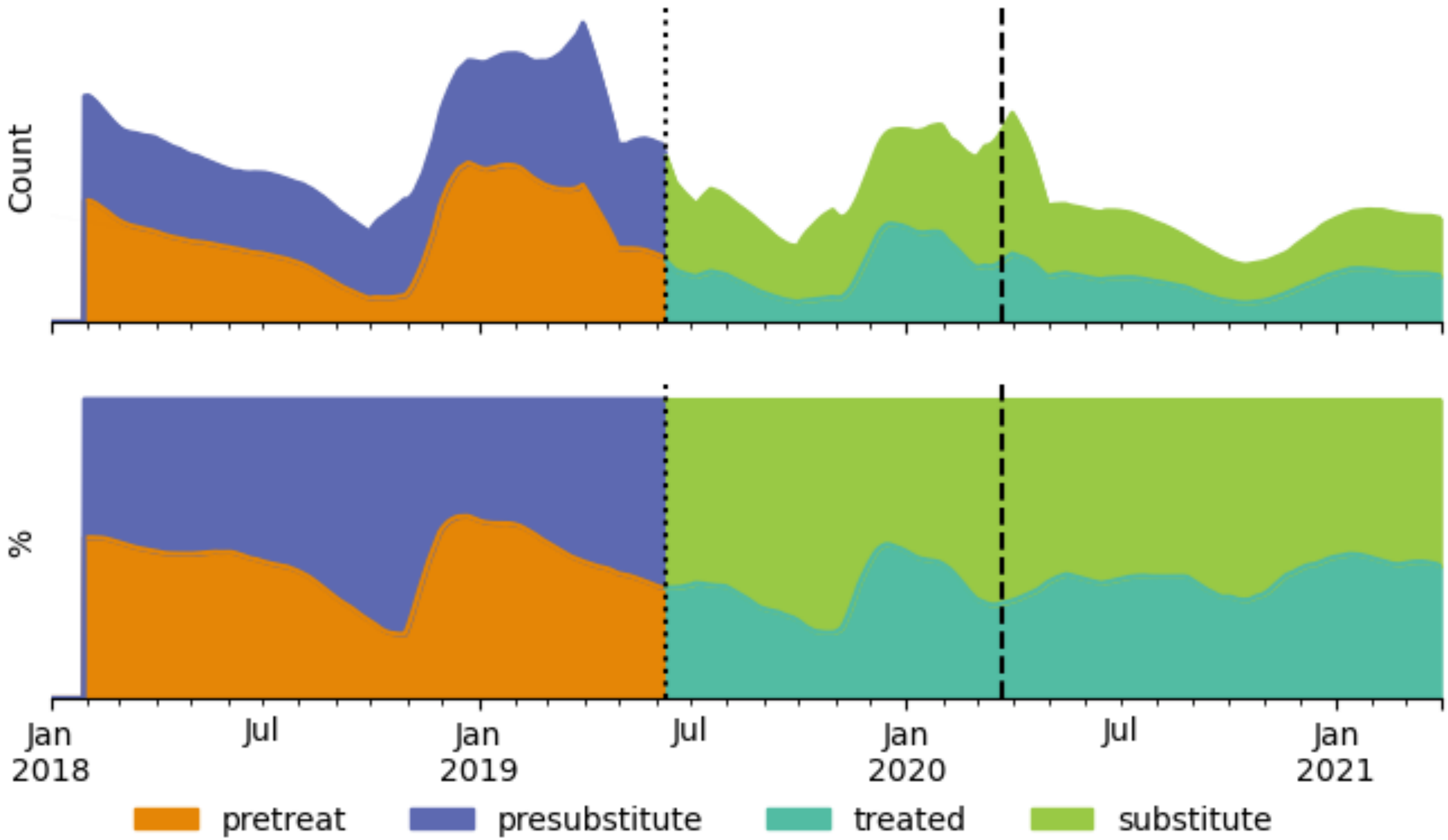
MATCH QUALITY



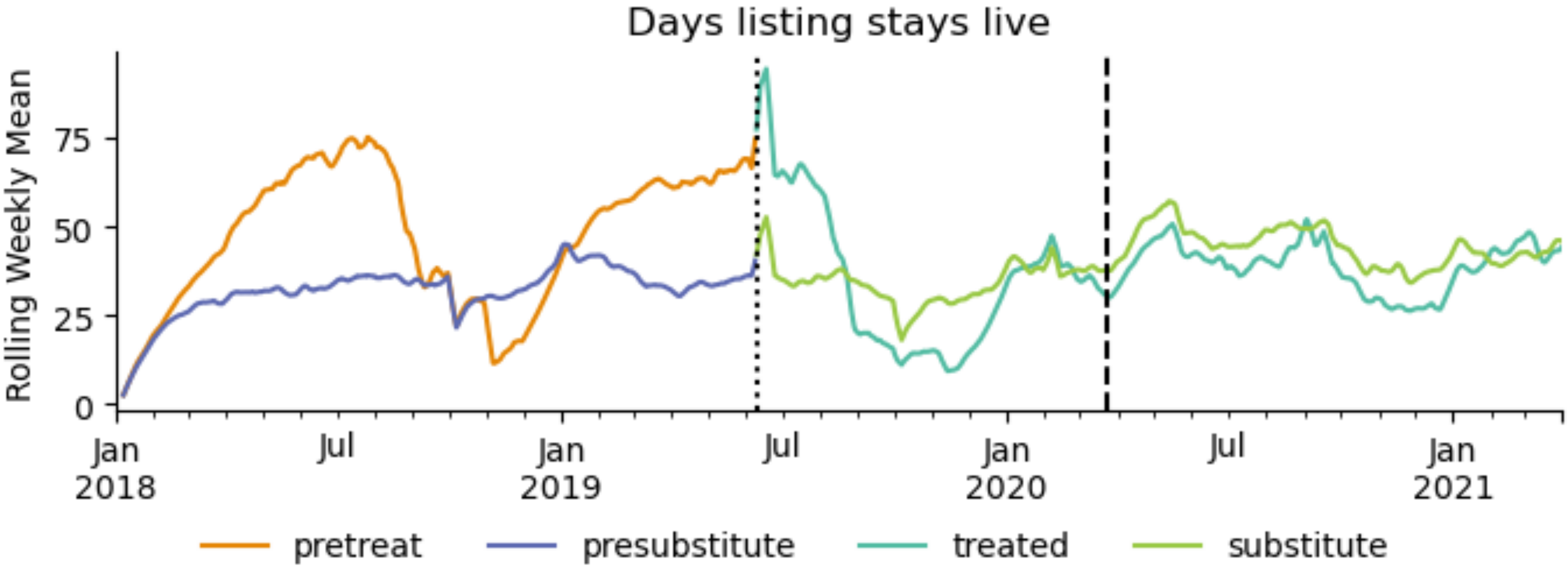
MATCHES ARE VERY BALANCED

Quartet A	Quartet B	KS Stat	Variable
treated	pretreat	0.11	Price
treated	substitute	0.25	Price
treated	presubstitute	0.34	Price
pretreat	substitute	0.15	Price
pretreat	presubstitute	0.26	Price
substitute	presubstitute	0.12	Price
treated	substitute	0.06	Listing Duration
treated	presubstitute	0.08	Listing Duration
pretreat	presubstitute	0.06	Listing Duration

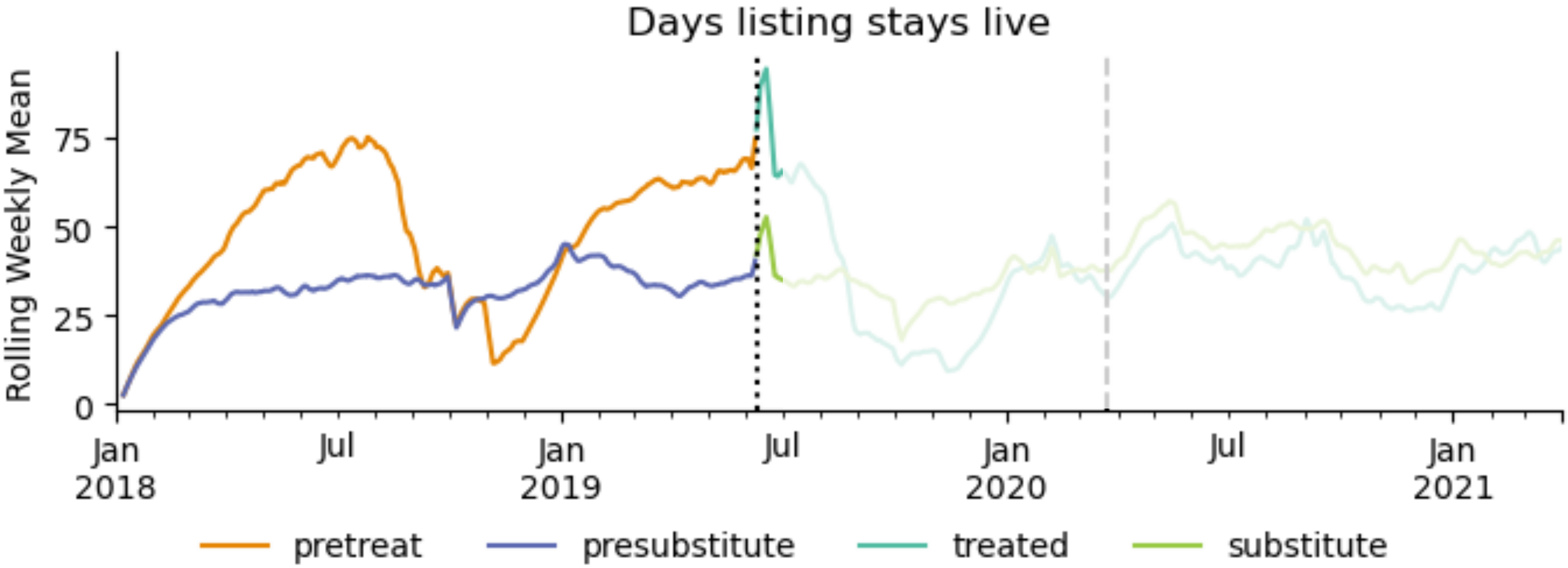
MATCHES ARE VERY BALANCED ON EXOG



SUPPLY EFFECTS: TOTAL LISTINGS



SUPPLY EFFECTS: STALENESS



Immediately after policy imposition, there's a bump that's not apparent in July 2018 or 2020, but…

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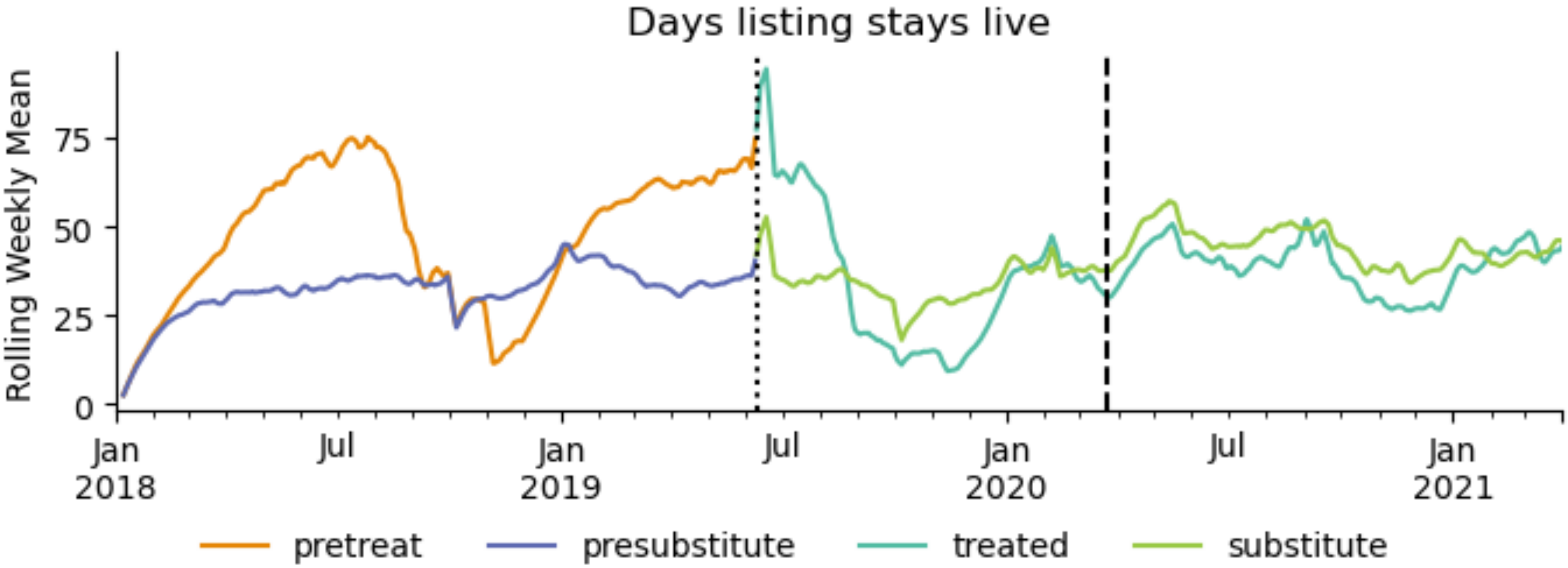

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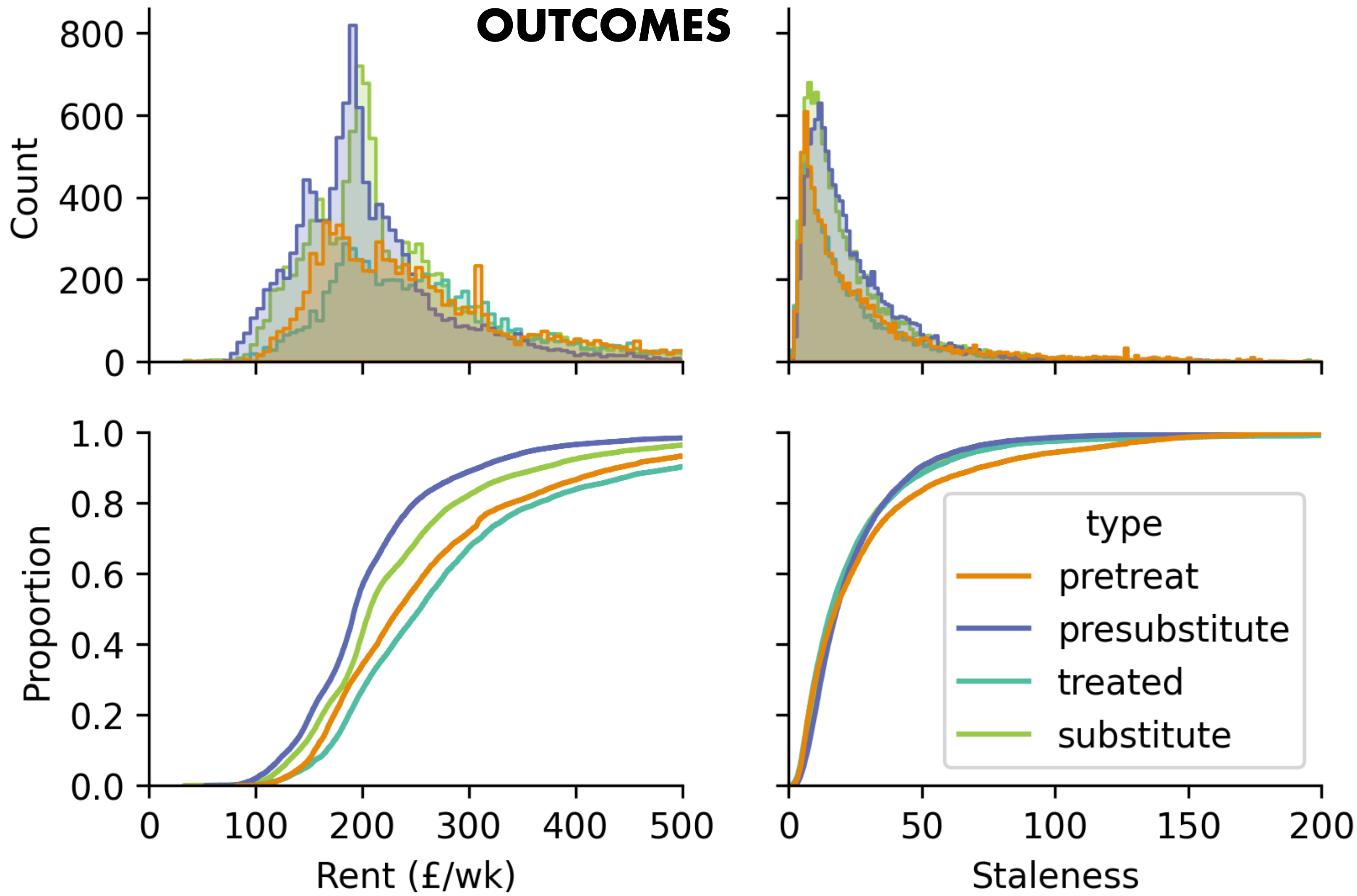
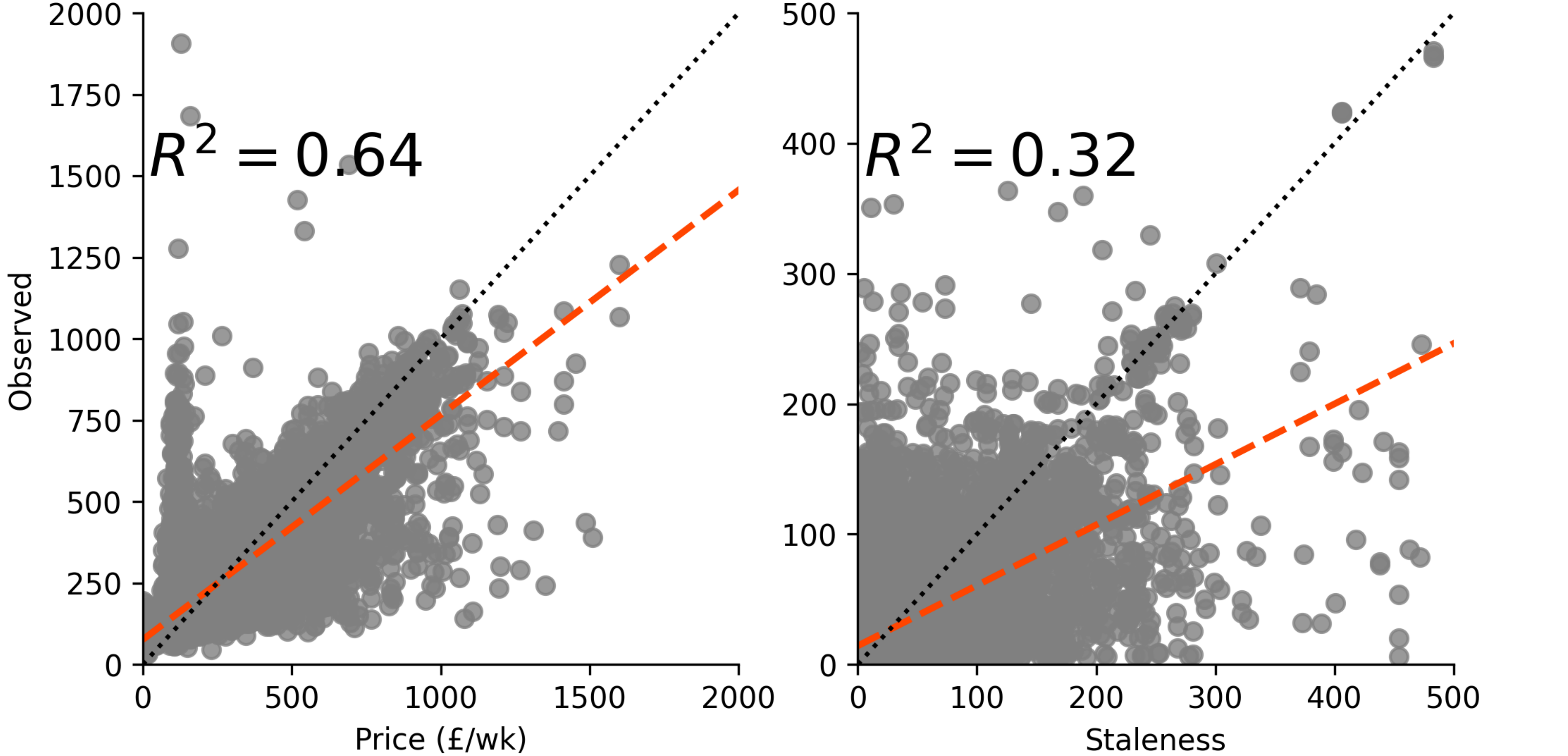


SPATIAL DIFFERENCE IN DISCONTINUITY



SPATIAL DIFFERENCE IN DISCONTINUITY

	Overall		Boundary	
outcome	duration	price	duration	price
presubstitute	-4.60	37.03	2.27	17.30
pretreat	-4.04	-26.15	0.38	-20.49
substitute	-2.51	-42.05	6.33	-7.98
treated	4.48	22.07	-0.83	14.63

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Difference between predicted pre-treatment price made using local models outside the DLA (actual) vs models inside the DLA (counterfactual)

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-25

35

SIZE OF THE GAP FLIP

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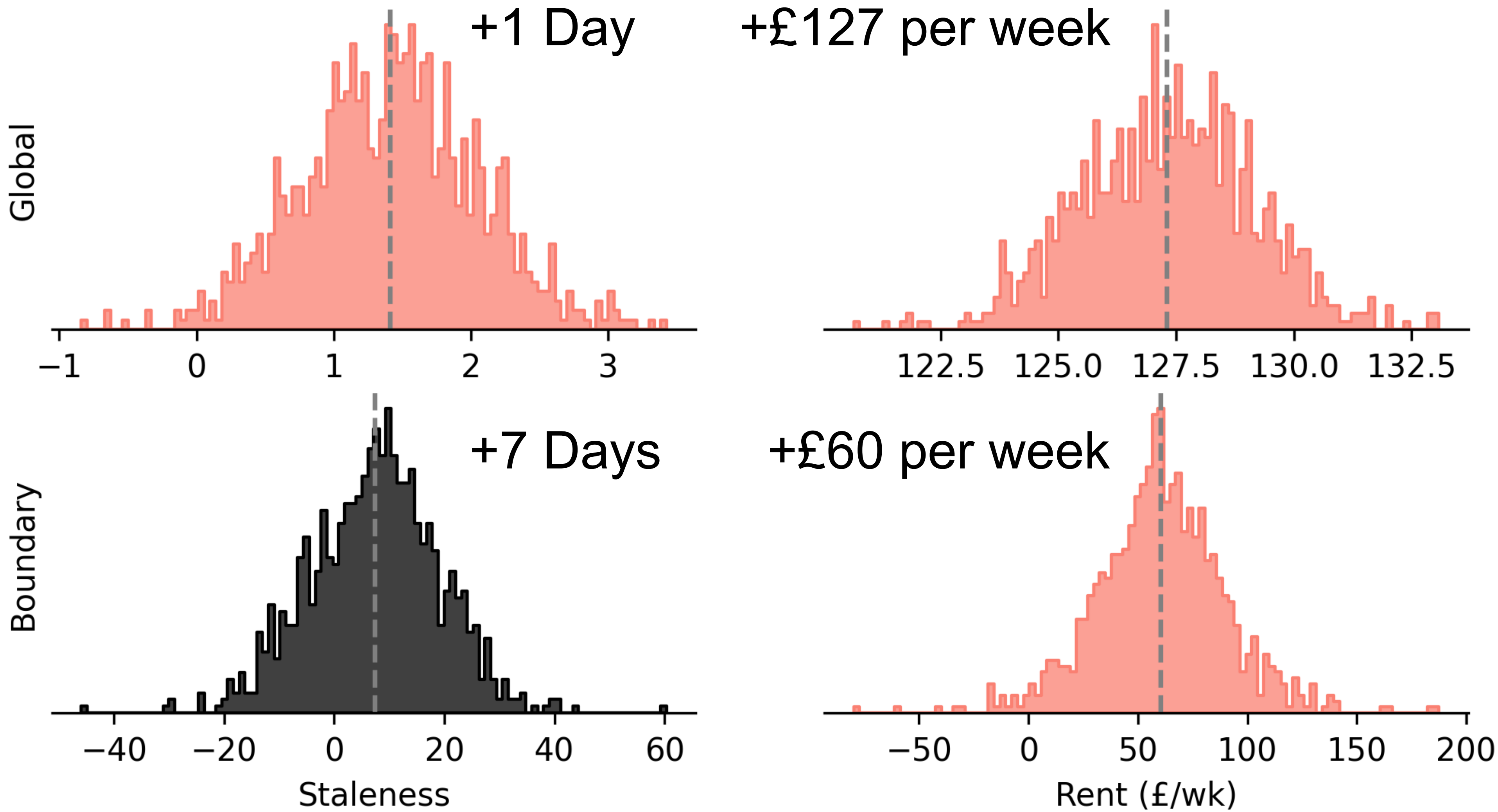
SDID: HIGHER PRICES, SLOWER CLEARING



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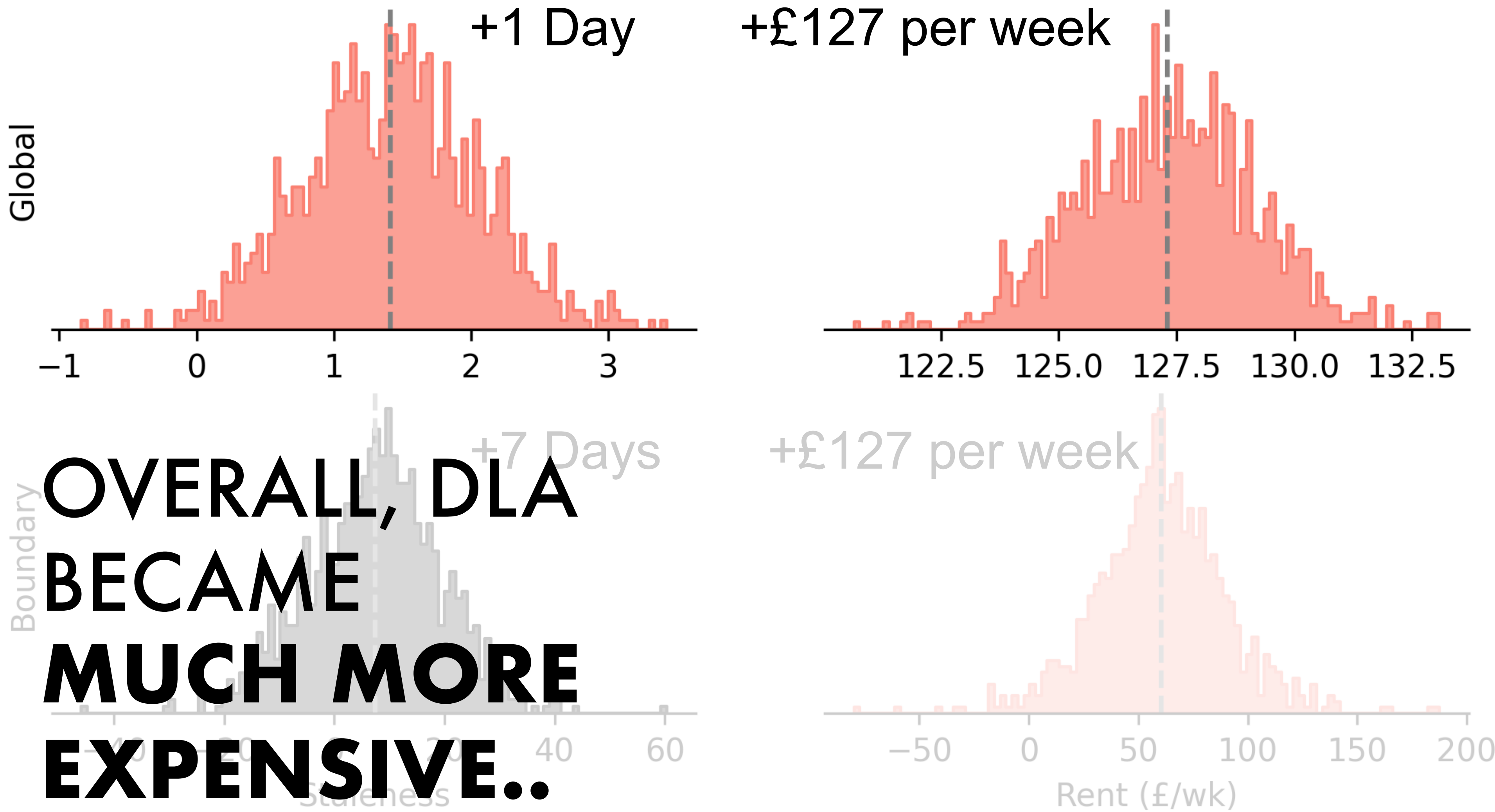

OVERALL, DLA BECAME **MUCH MORE EXPENSIVE..**

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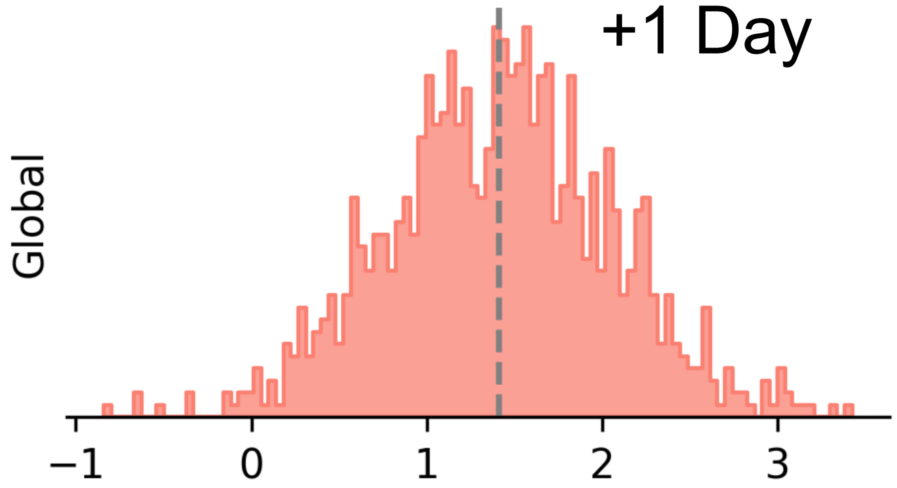

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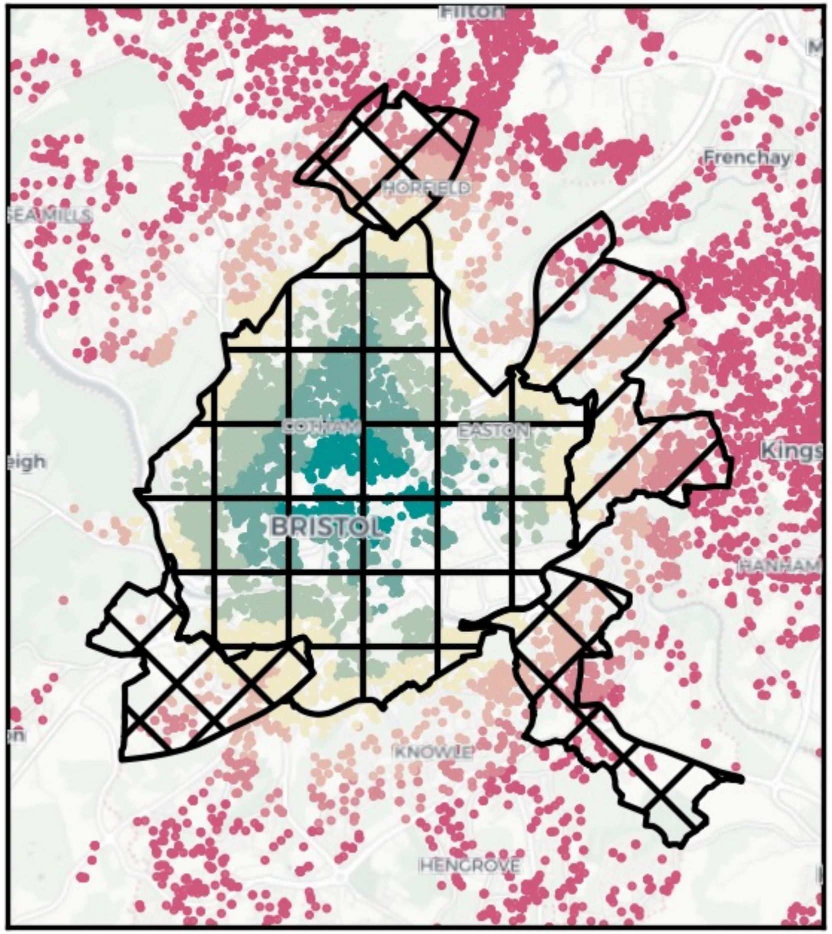
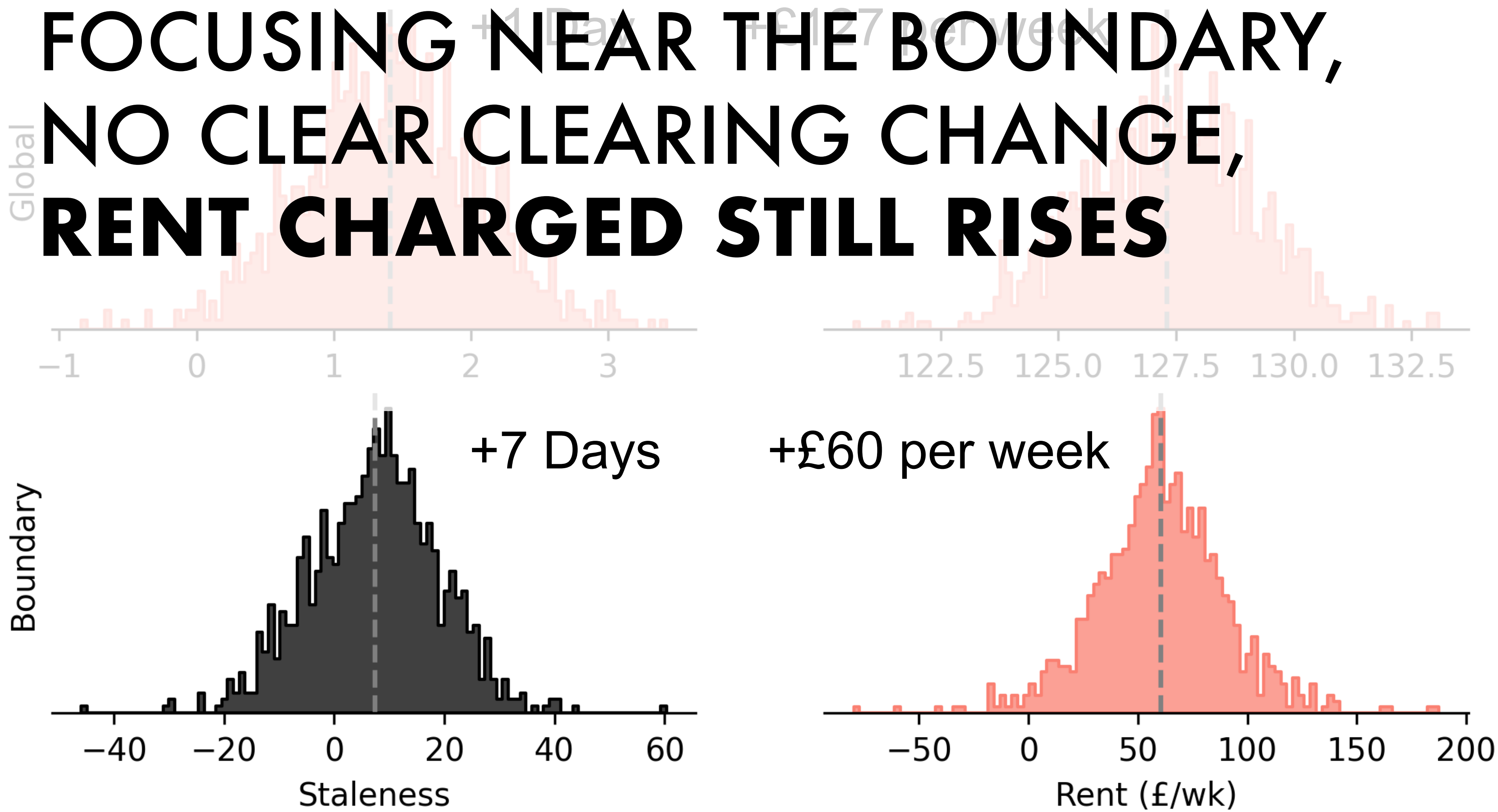
Listings near DLAs

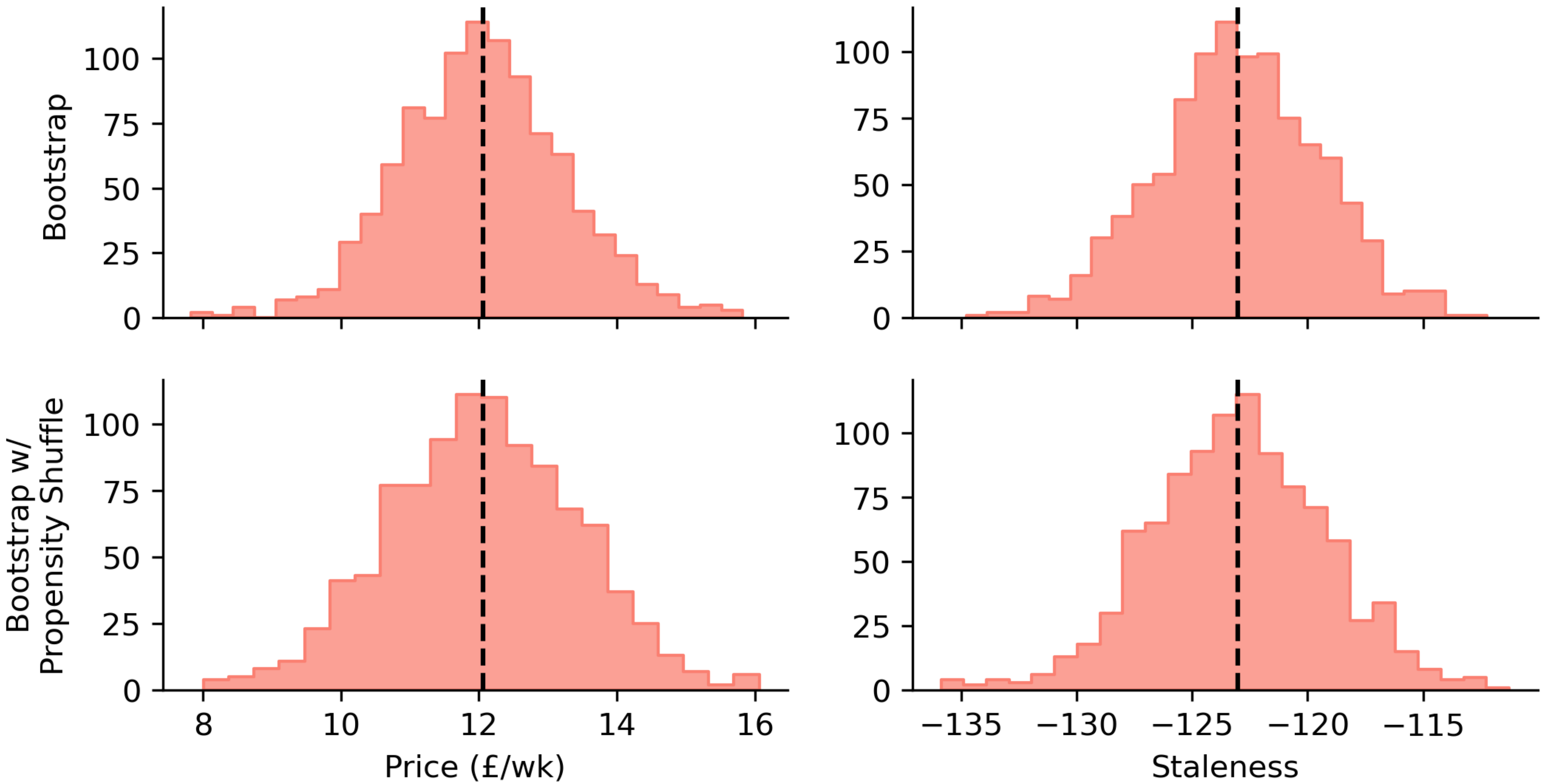


SPATIAL DIFFERENCE IN DISCONTINUITY

FOCUSING NEAR THE BOUNDARY, NO CLEAR CLEARING CHANGE, **RENT CHARGED STILL RISES**

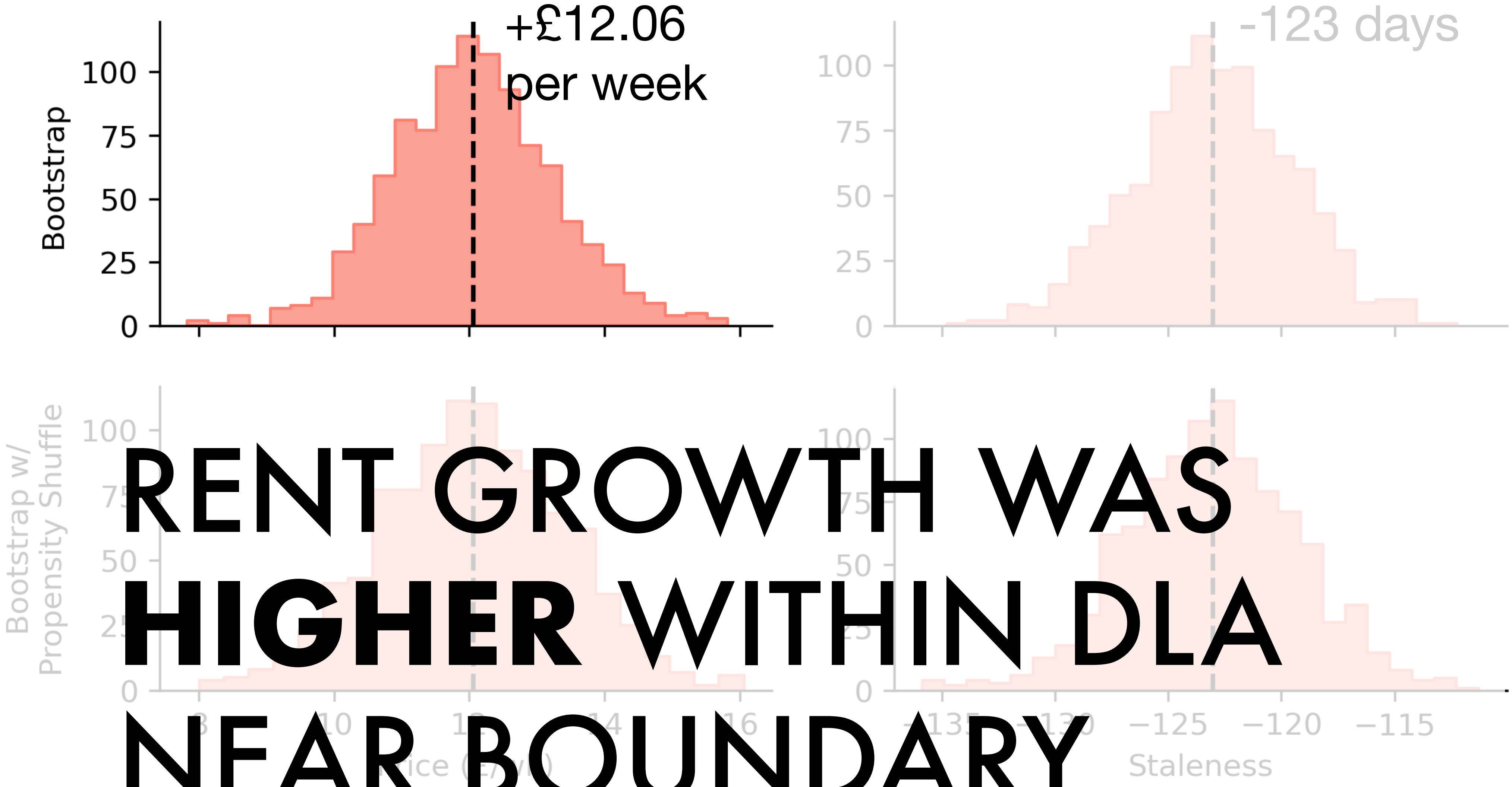


SPATIAL DIFFERENCE IN DISCONTINUITY



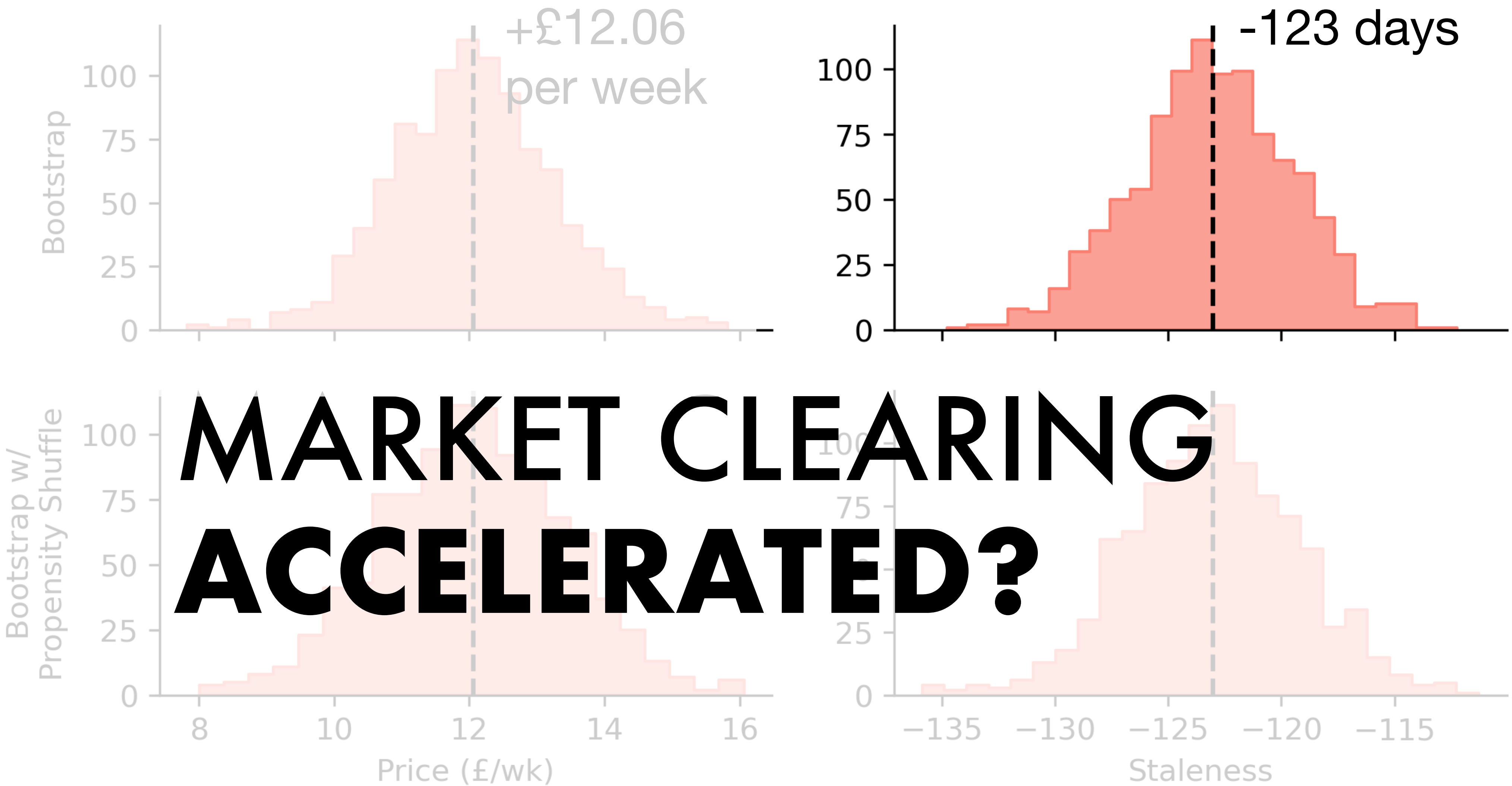
Bootstrap
Bootstrap w/ Propensity Shuffle
0
25
50
75
100
8
10
12
14
16
Price (£/wk)
−135
−130
−125
−120
−115
Staleness

XLEARNER



RENT GROWTH WAS **HIGHER** WITHIN DLA NEAR BOUNDARY

XLEARNER



MARKET CLEARING **ACCELERATED?**

XLEARNER

DLA Adoption caused faster rent growth, with an uncertain effect on clearing times for private rental properties, after controlling for COVID-19, stock imbalance, and path dependence.

Landlord concerns about fewer rentals & higher rent levels seem founded.

Need access to property register to determine partial treatment, new rental market entrants

SLA OR NAY?

SLA OR NAY?

CAUSAL IMPACTS OF BRISTOL DLAS

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Separate modelling into treatment models and control models. Each gets

$Y_j \sim Y_p + X_t + e_j$

Compute gap between observed and cfact prediction. For instance, if j is treated, calculate $\tau = Y_j - \hat{Y}_j$ with prediction from untreated model.

Estimate τ/ITE models separately again

XLEARNER DETAILS (1)

Estimate propensity to be treated p_j using

$T_j \sim X_j + Y_j$

Calculate propensity-weighted average treatment:

$ATE = \Sigma\, (\tau_u\, (1 - p_j) + \tau_t\, p_j\,)$

τ_t is the imputed ITE assuming individual j is treated, while τ_u is imputed ITE assuming not.

XLEARNER DETAILS (2)